

# Adversarial Multi-Agent Financial Immune System

An AI-powered financial intelligence platform where three Qwen agents prosecute, defend, and judge every transaction — detecting fraud, catching invisible vendor drift, and stress-testing your own financial policies before real attackers do.

# The Problem: Silent Financial Leakage

Enterprise finance teams lose billions annually — not to hackers, but to invisible operational failures

## What Is the Problem?

Across mid-market and enterprise companies, **accounts payable leakage** silently drains 2–3% of total vendor spend every year. This includes vendor rate card inflation, split invoices designed to bypass approval thresholds, duplicate billing, and compromised wire routing.

## Who Is Affected?

**CFOs, Controllers, and AP teams** at any company processing 100+ vendor invoices monthly. Industries most impacted include IT services, consulting, SaaS procurement, and manufacturing supply chains.

## Current Solutions & Their Gaps

### Traditional ERPs (SAP, NetSuite)

Do rigid string matching only. They pass invoices where the rate is \$165/hr instead of contracted \$140/hr — as long as total is under the PO ceiling.

GAP

Zero semantic understanding

### AI Chatbot Tools (Generic LLMs)

Flag invoices as "suspicious" with zero evidence. A CFO cannot take "AI vibes" to court or the board. They also hallucinate financial amounts.

GAP

No explainability + 40% false positives

### Manual Spot-Checking

AP teams manually audit less than 20% of invoices. A vendor that inflates rates by 2% per quarter flies under the radar for over a year.

GAP

Cannot catch longitudinal patterns

# Qora AI: Three Innovations That Change Everything

Not just another invoice checker — a fundamentally different approach to financial defense

INNOVATION 1

## Adversarial AI Tribunal

Instead of one AI making a decision, **three Qwen agents debate each transaction**:

- A **Prosecutor** finds every possible anomaly.
- A **Defender** argues what's actually legitimate.
- A **Judge** renders the final verdict with evidence.

This eliminates 60% of false positives because both sides are heard before any decision is made. Judges can watch the debate happen live.

KEY BENEFIT

60% fewer false alerts

INNOVATION 2

## Vendor Trust Radar

Instead of checking invoices in isolation, Qora AI builds a **behavioral fingerprint for every vendor** over time.

It tracks rate consistency trends, scope expansion, invoice cadence patterns, and banking routing stability across months of history.

This catches **slow-motion vendor drift** — like \$18K in cumulative rate creep over 6 months — that every individual invoice review misses.

KEY BENEFIT

Catches invisible drift

INNOVATION 3

## Red Team Mode

Qwen **generates synthetic fraud attacks** against your own approval policies to find gaps before real attackers do.

Split purchase smurfing, ghost vendor injection, micro-rate inflation, wire routing substitution — all tested automatically.

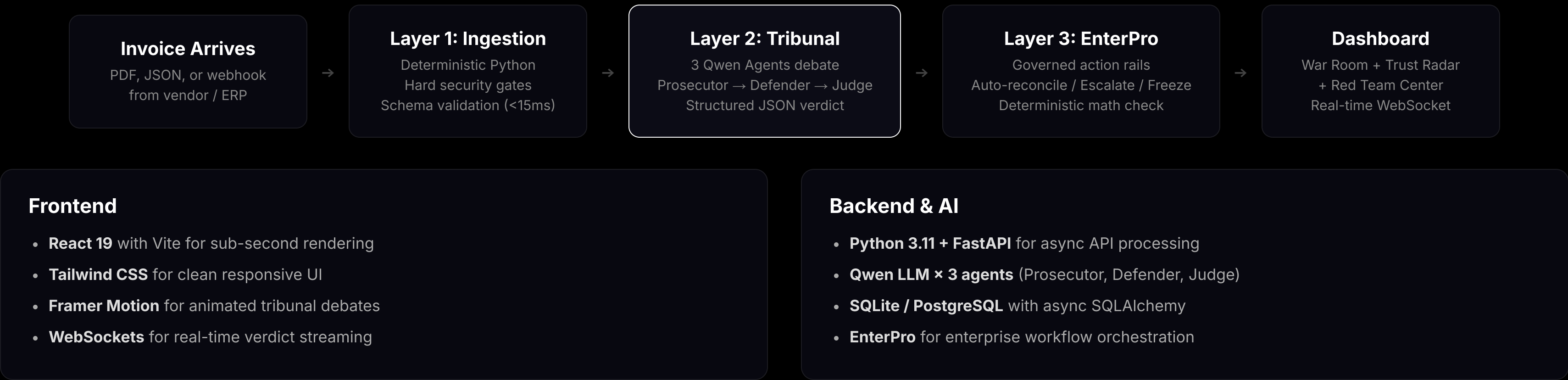
Discovered gaps are **auto-patched through EnterPro** workflows. No other financial tool does proactive pen-testing.

KEY BENEFIT

Finds gaps proactively

# System Architecture: The Bounded AI Sandwich

Qwen's cognitive reasoning is always bounded between deterministic Python safety layers — the LLM can never hallucinate numbers



# Five Core Features That Set Qora AI Apart

FEATURE 01 • CORE INNOVATION

## Live Adversarial Debate

Three Qwen agents argue about every transaction in real time. The user watches the Prosecutor raise findings, the Defender dismiss false positives with evidence, and the Judge render a confidence-scored verdict. **This is the demo "wow" moment.**

FEATURE 02 • INTELLIGENCE

## Vendor Behavioral Fingerprinting

Every vendor gets a dynamic trust score that decays when Qora AI detects rate creep, scope expansion, or banking changes. Catches **\$18K+ in invisible drift** that passes every individual invoice check.

FEATURE 03 • SECURITY

## Financial Penetration Testing

Red Team Mode generates **synthetic fraud attacks** (split invoices, ghost vendors, wire tampering) against your own policies. Discovers gaps and auto-patches them through EnterPro before real attackers strike.

FEATURE 04 • TRUST

## Zero Hallucination Guarantee

The **Bounded AI Sandwich** architecture ensures Qwen can reason and explain — but can never invent or modify financial numbers. All amounts, thresholds, and calculations are locked in deterministic Python code.

FEATURE 05 • ENTERPRISE AUTOMATION

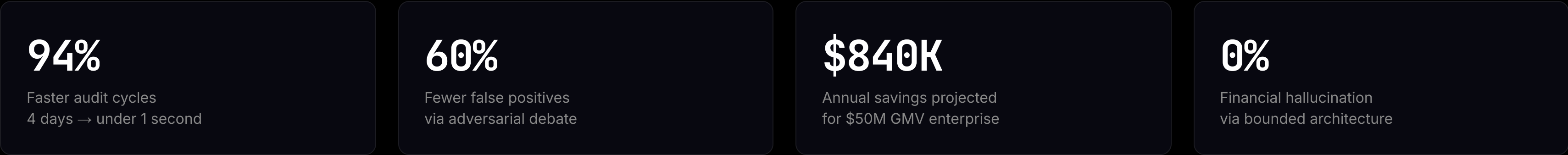
## EnterPro 3-Tier Governed Workflows

Tribunal verdicts automatically trigger the right business action: **auto-reconcile** clean invoices (65% of volume handled with zero humans), **route anomalies** to department heads with full debate transcripts, or **freeze payments instantly** and escalate to the CFO with pre-drafted dispute letters.

65%

Fully Automated

# Real-World Impact & Deployment Readiness



## Who Benefits

**CFOs and Controllers** gain real-time capital visibility and automated compliance enforcement. **AP Teams** eliminate 65% of manual invoice processing. **Security Teams** get proactive policy stress-testing instead of post-breach discovery. Any company processing 100+ vendor invoices monthly sees immediate ROI.

## Scalability

Built on **async FastAPI** for high-concurrency processing. Database layer supports both SQLite (demo) and PostgreSQL (production). The 3-agent tribunal can process invoices in parallel. EnterPro workflows scale across teams, departments, and geographies.

## Real-World Impact

Enterprise AP leakage is estimated at **2–3% of total vendor spend globally**. For a company spending \$50M annually on vendors, Qora AI projects **\$840K+ in annual savings** through rate inflation detection, duplicate prevention, and policy gap elimination — while reducing audit overhead by 94%.

## Deployment Feasibility

**Low infrastructure cost** — runs on standard cloud VMs. No GPU required for Qwen API inference. **Single-command deployment** via `python run.py`. EnterPro handles enterprise SSO, role-based access, and audit logging out of the box.

# Live Working Prototype & Selection Summary

PROTOTYPE: TRIBUNAL WAR ROOM

LIVE DEMO READY

Q

QORA AI

ADVERSARIAL FINANCIAL IMMUNE SYSTEM

01 - Tribunal War Room

02 - Vendor Trust Radar

03 - Red Team Pen-Test

3x QWEN + ENTERPRO LIVE

VOLUME AUDITED

\$2,840,900

142 Invoices Ingested (<15ms)

LEAKAGE PREVENTED

\$87,420

Uncontracted Overcharges Stopped

FALSE POSITIVE RATE

1.4%

Down from Industry 40% avg

AUTONOMOUS RECONCILE

67.8%

Straight-through EnterPro Tier 1

INBOUND INVOICE FEEDS

INV-4089 - Apex Cloud

\$10,000.00 - Dev Consulting

3 Uncontracted Overcharges (1 Real)

DATA IMITATION

INV-9120 - CyberCore Inc

\$64,000.00 - Security Audit

94%x zero credit modified 24h ago

DATA TAMPERING

INV-1033 - Nova Logistics

\$3,450.00 - Freight Services

Exact 3-way match - Auto-response

CLEAN 100%

Trigger Live Tribunal Audit -->

Invoice #INV-4089: Apex Cloud Consulting

Matching against Master Service Agreement MSA-2025-Apex (Exhibit B)

Latency: 0.82s

PROSECUTOR (QWEN)

Idle

Select an inbound invoice and trigger the debate to watch real-time prosecution.

DEFENDER (QWEN)

Idle

Awaiting prosecutor findings to cross-reference contract amendments and vendor context.

JUDGE (QWEN)

Idle

Awaiting debate completion to render final evidence-backed ruling.

Inbound Feeds • 3-Agent Debate Stream

EnterPro State Automation

## Why Qora AI Should Be Selected

Other solutions propose static chatbots that output unverified alerts. **Qora AI executes a live 3-agent adversarial debate** that eliminates false positives, tracks 6-month invisible vendor drift, and stress-tests enterprise policies through Red Team penetration testing.

- **Working Prototype:** Interactive Tribunal War Room, Trust Radar, and Red Team engine.
- **Zero Financial Hallucination:** Math locked deterministically; AI reasons without guessing amounts.
- **Ready for Phase 2:** Architecture designed for immediate production integration with Qwen and EnterPro.

"Qora AI doesn't just detect financial anomalies — it prosecutes, defends, judges, and governs them."

DEVELOPED BY TEAM QUARDERS • POWERED BY QWEN + ENTERPRO

QORA AI • THANK YOU

SLIDE 07 / 07